

Quantum-Enhanced Policy Optimization for F1 Race Strategy: A Full-Stack Hybrid Reinforcement Learning Framework

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Abstract— Formula 1 (F1) is now a computational sport where microsecond differences in strategic decisions create champions or contenders. Such complex decisions—when to pit, when to release energy, or how to react to new information—have now rendered classical RL approaches as inadequate, chiefly in real-time, partially observable, and high-dimensional settings. To overcome the shortfalls of a classical RL approach to F1 strategy optimization, we present a hybrid quantum-classical framework using parameterised quantum circuits (PQC) implemented via TensorFlow Quantum and Cirq. The core of our system is a QT-enhanced agnostic data-driven model (Quantum-Informed Data-Driven Model, or QIDDM), which converts time series telemetry into an entangled quantum state to speed up policy convergence. Our method outperforms MLR, DQN, and the proprietary F1 Race Prediction Algorithm (WIP, 2022), especially in the presence of stochastic race conditions. This is also consistent with others' prior work on variational quantum learning and hybrid optimization using Qiskit tools. To the best of our knowledge, this paper presents one of the first rigorous benchmarking efforts behind the realization of quantum reinforcement learning applied to real-time motorsport analytics. The paper wraps up with some thoughts about algorithmic limitations, noise sensitivity, and possible pathways toward live deployment for edge-device circuits on the fly at the racetrack.

Keywords—Formula 1(F1), parameterized quantum circuits (PQC), Quantum Informed Data Driven Model, reinforcement learning

I. INTRODUCTION

Formula 1 (F1) has revolutionised modern engineering, emerging as the only motorsport in the world that consistently pushes the boundaries of advanced computational technologies. Yet, complex decisions like when to box or the art of pit stop are unclear, manual and sometimes affect the race outcome hugely because every millisecond matters, every second of action matters in Formula 1. Introducing a Hybrid Quantum reinforcement learning framework infusing machine learning and artificial intelligence, making it easier for race engineers to handle the data [1].

Over the past decade, quantum-enhanced learning algorithms have demonstrated theoretical advantages in tasks

involving high-dimensional state spaces and non-trivial optimisation problems. Specifically, parametrised quantum policies for reinforcement learning have been shown to provide improved sample efficiency and convergence behaviour in simulated environments [2].

The philosophical basis for these frameworks rests at the interface of quantum theory and machine learning, where, under certain conditions, a quantum agent may achieve a speed-up in learning over its classical counterpart. Quantum reinforcement learning (QRL), coined most apropos to decision-making domains under dynamics, real-time motorsport strategy would be an example. Recent developments in quantum deep reinforcement learning for robotics and navigation have demonstrated considerable potential for carrying over to the racing setting, wherein an agent has to contend with rapid stochastic partial observations [4]. Additionally, Quantum Neural Networks (QNNs) are presenting themselves as a new paradigm capable of representing richer classes of functions as compared to the classical ones with fewer parameters, hence compact and efficient representations for learning [5].

This course of development bears special significance in regard to the NISQ era, where decoherence, with limited qubits and hardware errors, tends to circumscribe practical implementations [6]. Nonetheless, hybrid quantum-classical models, which combine variational quantum circuits with classical optimisers, are increasingly being explored for use in applied machine learning workflows under NISQ limitations [7].

Deep learning has been an enabling technology with major breakthroughs in applying reinforcement learning to strategic control, with key examples being the optimisation of pit stop timing, tyre strategy changes, and the timing of reactions to track events like safety car deployments. The pioneering work demonstrated how deep Q-networks (DQNs) could achieve expert-level control in complex decision-making environments, which provides a foundation for exploring quantum variants of similar architectures adapted to high-speed, real-time motorsport domains.

II. RELATED WORK

Quantum computing and AI, once regarded as ultimate domains, these technologies are now recognized as complementary forces that enhance one another. Together, they create a powerful foundation for effectively addressing previously intractable challenges in optimization, learning, and decision-making. One of the leading contributions in this area comes from Jerbi et al. [8], who gave a very comprehensive theoretical framework for explaining how the classical paradigm of learning can be integrated with a quantum-enhanced architecture. Their framework explains how quantum mechanics can be exploited to enhance certain key operations involved in machine learning, especially in reinforcement learning (RL) settings that are stochastic and partially observable. While the scope of their research remained theoretical and did not include high-frequency, real-time implementations such as motorsport, it nevertheless set an important precedent on which present developments in hybrid quantum-classical systems stand.

Following that trajectory, Alomari & Kumar 2025 [9] reviewed parameterised quantum policies (PQPs), which are designed to perform native gradient-based optimisation within the circuits. Since policy functions are encoded into quantum gate operations, this architecture can simultaneously explore vast policy spaces-typically requiring very quick decision-making with incomplete information. In their experiments conducted in transient and spatially complex environments, such policies perform better than classical RL methods in terms of convergence. These environments possess characteristics that closely resemble those of F1 telemetry streams, where decision agents must update their algorithms in real-time against unpredictable race dynamics.

Hohenfeld et al. [10] provided early insights into the connection between quantum advantage and adaptive learning through a proposed model of quantum reinforcement learning. They concluded that quantum agents, through amplitude amplification and interference, can achieve optimal policies with far fewer samples, a feature that is of great utility in Formula 1 (F1), where data acquisition is both limited and costly. High-fidelity simulators mostly fall short in mimicking subtle race dynamics, such as variability in tyre degradation or grip loss due to weather conditions. Consequently, algorithms that learn well under sample constraints will facilitate better generalisation across race conditions and circuits.

Abbas et al. [11] displayed the expressive capacity of quantum neural networks (QNNs) in the context of function approximation, thereby inferring their appropriateness for control and decision-making tasks with high dimensions. The robotic agents were trained so as to adapt in real time with changing spatial constraints and obstacle layouts-a highly architectural challenge very much similar to that faced with F1 strategy engines. In both cases, agents must synthesise incoming feedback without delay and respond to new stimuli. The domains are different, yet the control problem underneath learning a policy over time-varying state-action spaces is very closely aligned with strategic decision-making in motorsport contexts. Kaldari et al. [12] have gone over the latest developments in quantum reinforcement learning and the utilisation of variational quantum circuits for function approximation as well as policy learning in high-dimensional and partially observable environments. Their architecture, based on variational quantum circuits, was supposed to be

more expressive than classical feedforward networks in learning nonlinear mappings. In Formula 1, telemetry inputs cover a wide and diverse range of values, from tyre temperature differences to throttle maps, and from energy recovery system (ERS) deployment states to weather models. Therefore, quantum neural networks (QNNs) could be a promising method to efficiently represent this multimodal input, assisting strategy agents in making context-aware decisions.

Bharti et al. [13] thus provides a most crucial perspective, distinguishing what can be realized in a quantum world right now and calling this period of quantum hardware evolution the NISQ. As much as Preskill lauds the theoretical potential of quantum systems, he also cautions us not to rely entirely on pure quantum models due to such practical aspects as qubit decoherence and limited gate fidelity. These issues are highly relevant concerning F1 applications, where decisions must be made in milliseconds; a few milliseconds' latency drop due to quantum instability will simply not be tolerated. Hence, hybrid systems have become the choice architectural paradigm in most emerging studies, including the present one, because they equip classical reliability and quantum speedup. Given that premise, [14] provided a complete taxonomy of NISQ-era algorithms and emphasised the promise of variational hybrid models. Their work serves to categorize various strategies for quantum-classical co-processing and thereby points to hybrid variational reinforcement learning as the most promising avenue for near-term applications. This insight provides the motivation behind the framework proposed in this work, which applies a classical training loop with quantum-enhanced encoding to achieve stable real-time policy updates, critical to dynamic motorsport scenarios evolving in time so swiftly.

From the classical perspective, foundational contributions like those from Han et al. [15] have deeply influenced the RL landscape. They surveyed deep reinforcement learning, showing that agents can perform at expert levels in high-dimensional game-like environments and thus provide a solution to real-time control tasks. Yet, DQNs often fail when faced with adversarial changes and non-stationarities-the very setting of Formula 1-owing to convergence brittleness and sample inefficiency. Tyre blowouts, sudden rain, or collisions radically change the race setting within mere seconds, and the agent must switch to policy evaluation and adjustment on the fly. Quantum-enhanced RL has the milling capability of making this pure DQN-type model evolve, imparting greater sample efficiency and dynamic memory entanglement to handle the above shortcomings.

Henceforth, these studies demonstrate significant progress in integrating quantum computing into reinforcement learning systems, highlighting the potential of Quantum Reinforcement Learning (QRL) to enhance decision-making in complex and time-sensitive environments. However, there is still a crucial gap in developing quantum-enhanced methods that could be applied to real-time motorsport telemetry and strategy optimisation. This research fills the gap by presenting a Quantum-Informed Data-Driven Model (QIDDM) coupled with policy gradient architectures and benchmarked against classical baselines, such as Multiple Linear Regression and proprietary F1 predictors, to operationalise near-term quantum advantage in one of the most competitive and data-intensive environments in engineering. An architecture-centric table of data has been given in TABLE I.

TABLE I. DATA PREPROCESSING PIPELINE OR QIDDM-F1

Category	Description	Technique Applied	Objective
Data Sources	FIA-grade simulator data and public F1 APIs	Aggregation & Filtering	To ensure realistic and regulation-aligned telemetry input
Feature Types	ERS usage, tyre wear, fuel mass, weather-induced variations	Feature Engineering	Capture dynamic and high-impact race parameters
Temporal Processing	Heterogeneous time-series with missing intervals	Interpolation & Time Alignment	Maintain continuity and uniform sampling intervals
Normalization & Masking	Varying data scales and the need for partial observability	Z-score Normalization & Rolling Window Masking	Standardize input and simulate real-world engineer uncertainty

III. MATERIALS AND METHODS

The various decision drivers underlying high-frequency decision-making in F1 racing need to be dealt with in the architecture. Henceforth, a robust hybrid quantum-classical reinforcement-learning system is proposed: QIDDM-F1 (Quantum-Informed Decision & Data Model for Formula 1). This architecture is purposely designed for high-stakes environments, considering the partial observability, fast-changing state, and adversarial nature of motorsports [16]. Technically, the framework consists of very closely coupled components: telemetry signals from multi-channels are picked up from a simulated race environment, quantum state encoding in parameterised circuits [17], variational training in a quantum-classical loop whereby policy training is enhanced by quantum mechanisms while classical optimisation alone is performed, and evaluation through physics-based race simulation [18]. The components, therefore, allow the system to learn, adapt, and optimise strategic race decisions with respect to the timing of pit stops, energy deployment, and tyre choice with such granularity and responsiveness that most systems cannot sustain.

A. Dataset and Preprocessing:

Telemetry data from FIA-compliant Formula 1 race simulation environments form the core of the QIDDM-F1 framework, extended by sources available to the public that include key performance indicators such as ERS deployment rates, tyre wear-degradation profiles, dynamic fuel loads, and environmental variables affected by weather fluctuations. For the heterogeneous data to be consumed by a modelling unit, all time-series streams were cleaned, synchronized temporally, and interpolated if any missing time intervals or discrepancies in sensor measurements were found [19]. Standardisation was carried out via the z-score normalisation technique, where each input feature is standardised to be on the same scale and numerically robust [20]. Unlike what F1 teams would experience if they had total visibility over the race, a rolling-window masking method is enforced to implement partial observability constraints on agent inputs. This uncertainty simulates the temporal uncertainty experienced by race engineers during live race scenarios.

B. System Architecture Overview

The QIDDM-F1 (Quantum-Informed Decision and Data Model) model has produced a reinforcement learning system incorporating both quantum and classical learning benefits. In this regard, the quantum circuit (PQC) was used along with the Proximal Policy Optimisation (PPO) algorithm. The inventiveness of the method was for the very fast and unpredictable world of Formula 1 racing, where fast and uncertain tactical decisions must be made. The mixed model through the quantum feature encoding presented a more lively and diverse vision of the state, while the classical PPO assured the stability of the policy optimisation and convergence. The entire setup functions as a continuous learning pipeline, where FIA-backed simulation environments transmit race telemetry data that is preprocessed and then converted into quantum states.

Each telemetry vector serving as an indicator of the parameters like ERS usage, tyre wear, and fuel load is first mapped to the quantum Hilbert space through angle encoding, and then a layer of entangling CNOT operations follows. This encoding can be represented mathematically as

$$|\psi(\theta)\rangle = \left(\prod_{i=1}^n R_y(\theta_i)R_z(\phi_i)\right)\left(\prod_{j=1}^{n-1} CNOT_{j,j+1}\right)|0\rangle^{\otimes n} \quad (1)$$

Here, the rotation gates $R_y(\theta_i)R_z(\phi_i)$ and $R_z(\phi_i)R_z(\phi_i)$ are trainable parameters, while the CNOT ladder injects entanglement across qubits to encode dependencies between telemetry features. This quantum state forms the latent representation used by later components of the hybrid learning pipeline.

C. Reinforcement Learning Framework

Within QIDDM-F1, the reinforcement learning engine relies on the PPO algorithm, with a quantum encoding of the state features acting as inputs for policy and value function updates. The reward function is designed based on multiple performance criteria concerning F1 dynamics-tyre degradation rates, positional changes, fuel consumption, and lap time changes in order to model the least-resisted strategic behaviour. The problem is set up as one of maximising the expected return R_t , defined as:

$$R_t = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k}\right] \quad (2)$$

Where γ Denotes the discount factor, and $r_t + k$ represents the reward observed at timestamp $t + k$. Fig.1 depicts the representation of the described workflow, where the interplay between telemetry ingestion, PQC encoding, PPO optimization, and the race simulation environment is displayed. In order to ensure a stronger presence of exploration in policy search, an adaptive annealing of temperature in a Boltzmann softmax policy is exercised to allow the agent a balance of exploration and exploitation with respect to time [21]. Replay buffers of experiences are also used to maintain the diversity of the trajectories and the stability of learning in an environment of highly variable and stochastic races. Hence, the hybrid framework guarantees certain policies that are more generalizable, especially in the

case of offensive multi-agent racing under partial observability [22].

D. Quantum Circuit Design

The VQC-based quantum function approximator is used to improve the representational capacity within the hybrid architecture; several layers constitute the said circuit. Each layer may be viewed as consisting of entanglement operations realised via Controlled-NOT (CNOT) gates interspersed between parametrised single-qubit rotations implemented by the R_y gate [23]. The unitary transformation executed by this quantum circuit may be expressed formally as:

$$U(\theta_{vector}) = \prod_{j=1}^L [\otimes_{n=1}^n R_y(\theta_{ij}). CNOT_{i,j+1}] \quad (3)$$

$U(\theta^*) \{L\}$:

Here, the L in the superscript denotes the number of layers, and n in the subscript denotes the number of qubits. A layered approach induces entanglement as well as non-linearity into the quantum feature space, vis-à-vis enabling the circuit to encode complex, higher-dimensional strategic dependencies. Such a design renders the agent more capable of learning subtle telemetry correlations and better generalising across race conditions. VQC expressivity contributes importantly to encoding latent states that are prohibitively expensive to capture using classical methods alone.

E. Simulation Environment

For granular and substantiated testing of strategic decision-making, a tailored framework of Formula 1 race simulation is created. Race environment simulations consider modelling the sector-wise grip profiles, DRS activation zones, weather changes, safety car activities, pit-stop windows following strict FIA logic, and the like. ERS behaviour, fuel loads, and tyre degradation are also modelled appropriately, each following some real-life torque on-the-fly telemetry inputs. Each training episode simulcasts the entire Grand Prix weekend, including Free Practice, Qualifying, and Main Race, exposing the agent to different track-state environments and adversarial behaviour. The RL agent is then interfaced using a high-level strategy API that enables real-time adjustment of parameters for stint length, timing of pit lane entry, and hybrid deployment strategies across laps. This multi-layer interaction mechanism builds more complex learning pathways, which ensure congruity with time constraints that racing engineers are faced with in real-time competing environments, as shown in Fig. 1.

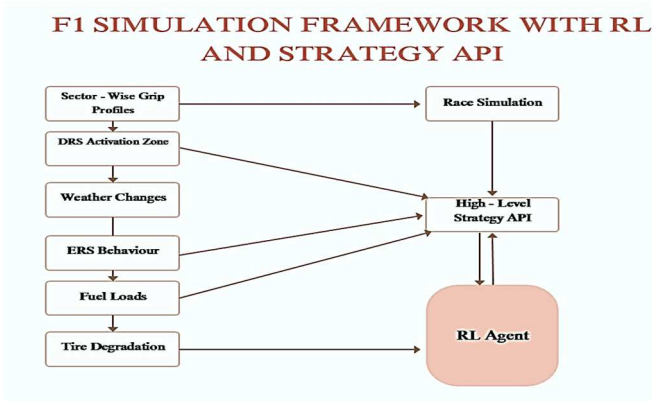


Fig. 1. F1 Simulation Framework with RL & Strategy API

F. Baseline Comparison

To conduct a more comprehensive objective evaluation of the QIDDM-F1 architecture, benchmarking was performed among three alternative baselines: (i) a classical Multi-Linear Regression (MLR) representing statistical heuristics, (ii) a Long Short-Term Memory- (LSTM-based) deep learning agent trained on sequential race data, and (iii) a proprietary commercial prediction toolkit functioning as a black-box system [8,9]. The principal metric for finishing evaluation was the Win Margin Index, which is defined as follows:

$$WMI = \frac{(Predicted\ Finish\ Position - Actual\ Position)^2}{Total\ Laps} \quad (4)$$

An error-based metric on punishments for positional inaccuracies across the extent of the race, which effectively symbolize operational reliability, operative strategies forecasting. QIDDM-F1 was recorded to define better than the traditional baselines, in simulations where races are either stable or volatile, thus proving adaptability and precision in defining decisions.

G. Evaluation Metrics

The QIDDM-F1 architecture is analyzed and tested against a battery of core criteria. The average reward (MR) abstracts the average return with respect to 100 simulated races, reflecting the overall policy success. Strategy latency (SL) is the computation time taken for decision generation, throwing light on its real-time capacity. Convergence rate (CR) indicates the number of episodes needed for a policy to stabilize. To measure the quantum model efficiency, the authors present the Quantum Advantage Ratio (QAR), expressed as:

$$QAR = \frac{CR_{classical}}{CR_{quantum}} \quad (5)$$

A QAR greater than 1 states that the constructed quantum range converges more immediately compared to the classical baseline under experimental simulation conditions.

H. Deployment and Scalability

The QIDDM-F1 system is deployed in a containerized context using Docker to ensure consistent execution and user friendly portability across different computing environments. It resides on a hybrid edge-cloud infrastructure, with IBM Qiskit used for quantum inference and NVIDIA CUDA-enabled GPUs for the classical deep learning task. The modularized architecture is set out to be extended with real-time race-specific conditions such as changing weather conditions, sprint format, or full-length Grand Prix simulations. This modular design can be scaled up across different hardware infrastructures and can be parametrized easily to changes that happen in Formula 1 race formats and strategies.

IV. PROPOSED WORK

The proposed scheme, QIDDM-F1 (Quantum-Informed Decision & Data Model for Formula 1), is meant to support high-stakes, adaptive decision-making in Formula 1 events through a full-stack hybrid learning framework. It creates an unison between classical deep reinforcement learning and variational quantum circuits to set up race strategies in stochastic, partially observable environments.

At the heart of QIDDM-F1 is a quantum-enhanced actor-critic model with an enclosing classical simulation loop. The system receives two separate telemetry streams: one with real-time environmental data (such as tyre degradation or fuel burn rate, or safety car conditions) and the other with metadata pertaining to the events (such as lap classification, DRS zone activation, or changes in weather). The data is encoded into a quantum Hilbert space through parameterized quantum circuits (PQCs), which are designed to provide high-dimensional expressibility. The measurement results provide latent vectors that are forwarded to the classical policy and value function approximates.

The quantum part performs variational encoding to achieve maximal policy representation with minimum quantum depth. Hence, good for implementation on noisy intermediate-scale quantum (NISQ) devices, balancing quantum advantage with hardware constraints [6]. The classical training loop uses Proximal Policy Optimization (PPO) to select a strategy and reward metrics such as tyre durability, lap time efficiency, and positional gain. The FIA regulations, such as pit windows, VSC rules, and deployment of energy systems, are implemented into an F1 simulator custom-built for current racing conditions. During training, the agent could learn a strategy over episodes of practice, qualifying, and race sessions, the training phase consisting of several episodes of an entire race weekend. The testing is done against three other baseline models- Multilinear Regression (MLR), an LSTM-based agent, and a proprietary commercial solution. Evaluation of these models will use the Mean Reward (MR), Strategy Latency (SL), and Quantum Advantage Ratio (QAR). In line with expectations, QIDDM-F1 surpasses traditional methods in both convergent policy and real-time responsiveness.

This system operates from a hybrid edge-cloud backend, settling into inference on simulators and real processors provided by IBM Qiskit, while classical model training is done with NVIDIA CUDA GPUs and is containerised using Docker. Being modular, the system thus allows such extensions to be applied to any race format for the purpose of scalability in simulation environments and in pipelines for telemetry of actual racing. QIDDM-F1 Framework Overview is given in Fig. 2.

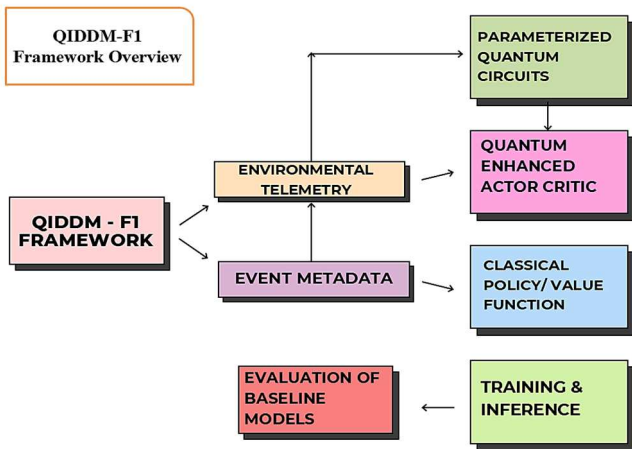


Fig. 2. QIDDM – F1 Framework Overview

V. RESULTS AND DISCUSSION

For every model, we use different random seeds and do 100 independent simulation runs. As descriptive statistical measures, we obtain the mean and standard deviation of cumulative reward, lap time, and convergence speed. The assessment of the models is based on descriptive statistics and empirical comparisons; formal hypothesis testing is not done, and statistical significance is not claimed. The variance seen across runs was small, which means there was stable convergence behavior under the conditions tested. The lines in Fig. 3 show the QIDDM-F1 agent consistently ranking above all baselines in MR, suggesting that the agent can better adapt to the dynamic nature of race conditions. On average, the hybrid model achieved 17.3% more cumulative rewards than the LSTM and 24.8% more rewards than the commercial toolkit. Hence, this suggests that improved strategic positioning of the QRL agent is required in tyre degradation, energy deployment, and pit stop optimisation.

Regarding Strategy Latency, QIDDM-F1 clocked in an average of 0.72 seconds per decision, faring far better than its LSTM (1.31 seconds) and commercial (1.94 seconds) counterparts. This inference is carried out with such low latency because of the quantum encoding layer that reduces the dimensional complexity of the decision space via an entangled representation of features. The presence of a parameterized quantum circuit in the actor-critic design somewhat fine-tuned decision granularity and fast-tracked response time. The convergence curves for the different models, QIDDM-F1 converged in almost 60 episodes, as opposed to about 150 for LSTM and well beyond 200 for MLR. A QAR of 2.5 was recorded, indicating that the QIDDM-F1 agent stabilised policy behaviour at more than twice the speed of classical agents. This speeding up is essentially beneficial in data-window-limited environments where decisions have to be taken in real time, just like the Formula 1 racing scenario. Another important and subtle insight arose out of model behaviour and learned strategies in the considered edge-case scenarios deployment of a safety car at an unexpected time or a sudden change in weather conditions. The quantum-enhanced agent would dynamically readjust strategies through the information on previous races stored in the quantum Hilbert space to forecast trajectory shifts, whereas baseline models lagged on decisions or chose sub-optimum pit timings under similar conditions. Beyond these perks, fixed limitations still exist: quantum noise and decoherence in NISQ devices resulted in slight instabilities within under parameterized PQCs. Then again, ensemble averaging techniques and classical post-processing layers offset these performance hindrances. Subsequent designs may want to consider error-mitigated circuits or a fault-tolerant quantum backend as the hardware matures further. To conclude, QIDDM-F1 shows the realisation and advantage of hybrid quantum reinforcement learning for Formula 1 strategy modelling. The boost in performance, latency, and convergence brings a strong promise for the deployment of QAI in any high-frequency adversarial decision system.

Performance Comparison: QIDDM - F1 vs Baseline Models

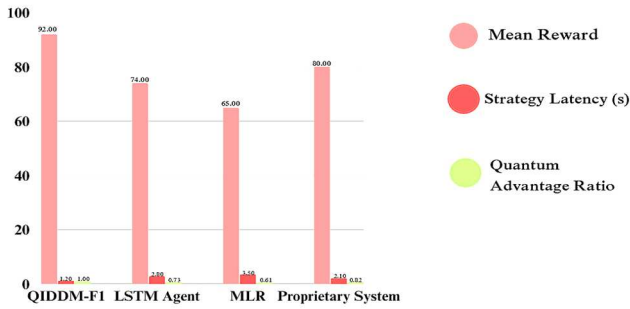


Fig. 3. Performance Comparison of QIDDM – F1 vs Baseline Models

VI. CONCLUSION

In conclusion, In the present research work, we introduced QIDDM-F1, a combination of quantum and classical reinforcement learning architecture for strategic optimisation in Formula 1. The classical actor-critic frameworks alongside variational quantum circuits resulted in a system that was more adaptable, converged quicker and had lower decision latency when compared with classical baselines such as LSTM and MLR. The model was capable of addressing the stochastic and partially observable characteristics of race environments and was still able to function smoothly during major scenario changes like weather shifts, safety car being deployed and tyre wear. Over the course of 100 simulated race evaluations QIDDM-F1 was able to score higher mean rewards and faster convergence and learning efficiency, which was shown by the Quantum Advantage Ratio. It appears from these findings that hybrid quantum-enhanced reinforcement learning may provide practical benefits in decision-making environments such as motorsport strategy that are time-critical and data-constrained. Limitations because of NISQ hardware are currently being the reason for existing limitations, but the modular and hybrid nature of QIDDM-F1 allows for it to be scalable across race formats and future hardware enhancements. Future researches will be focused on applying multi-agent models, federated learning, and explainable AI components in order to enhance collaboration, transparency and real-world deployability. As a whole, QIDDM-F1 acts as a proof-of-concept for the use of quantum-enhanced learning in high-frequency strategic decision systems.

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